03. CONDITIONAL AND LOOPS.

Conditions: It provide check for the statement.

01. If-else statement: Used to check the condition, it checks the boolean condition true or false.

Syntax:

```
if (boolean expression true or false) {
        // Body
} else {
        // Do this
}
```

Example :

```
Public class main {
    Public static void main (String[] args) {
      int age = 19;
      if (age >= 18) {
          System.out.print (" you are able to vote");
      } else {
          System.out.print (" you are not able to vote);
      }
    }
}
```

output

you are able to vote.

02. Multiple if-else Statement : It excutes one condition fuom multiple statements

Syntax

```
if (Condition 01){
    // Code to be executed if Condition 01 is true
} else if (Condition 02) {
    // Code to be executed if Condition 02 is true
} else if (Condition 03) {
    // Code to be executed if Condition 03 is true
} else {
    // Code to be executed if all Conditions are
        false.
}
```

Example:

```
Public Class Main {
    Public Static void main (String[] args) {
        int Salary = 25400;
        if (Salary <= 1000) {
            Salary += 1000;
        }
        else if (Salary <= 2000) {
            Salary += 2000;
        }
        else {
            Salary += 3000;
        }
        System.out.print(Salary);
    }
}
```

Output:

28400

LOOPS

Loops: Loops are used to iterate a part of program several iteams.

01. FOR loop: It is generally used when we know how many iteams loop will iterate.

Syntax:

```
for (initialization, Condition, increment/decrement){
    // body
}
```

Example

```
Public class Main {
  Public Static void main (String[] args) {
     for (int i=1 ; i<=5 ; i++) {
        System.out.print(i);
     }
  }
}
```

output: 1, 2, 3, 4, 5 (in downward)

02. While loop : It is used when we don't know how many time the loop will iterate.

Syntax :

```
while (Condition) {
    // Code to be executed.
    // increment / decrement
}
```

Example :

```
public class whileloop {
    public static void main (String [] args) {
        int num = 1
        while (num <= 5) {
            System.out.print (num);
            num += 1;
        }
    }
}
```

Output : 1, 2, 3, 4, 5 (in downward)

03. do while loop: It is used when we want to execute our statement at least one time.

It is called exit control loop because it checks the condition ~~one~~ after execution of statement.

Syntax:

```
do {
    // Code to be executed.
    // update statement → ++ / --
} while (condition);
```

Example

```
Public class Main {
    Public Static void main (String[] args) {
        int n = 1;
        do {
            System.out.Print(n);  n++;
        } while (n<=5);
    }
}
```

Output

```
1
2
3
4
5.
```

Comparision between while loop and do while loop

While Loop	Do While Loop.
01. Used when no. of iteration is not fixed.	01. Used when we want to execute the statement at least once
02. Entry Controlled loop.	02. Exit Controlled loop.
03. no Semicolon required at the end of while (Condition)	03. Semicolon is required at the end of the while (Condition);